SYLLABUS OF INFORMATICS ENGINEERING DEPARTMENT

Course Code	BAI20103J
Course Name	Artificial Intelligence (AI)
Credit Value	3 (3+0)
Course Classification	Core
Semester	2
Synopsis	Artificial Intelligence (AI) course is a field of computer knowledge that is intended to create software that can mimic human intelligence. This course describes methods for making software smarter, more logical and more useful for helping humans solve problems using artificial intelligence techniques. Intelligence techniques include search/tracking techniques, knowledge representation and logic, expert systems and soft computing, namely fuzzy logic, genetic algorithms, and artificial neural networks.
Course Learning Outcome (CLO)	At the end of the course, students will be able: CLO1: Student know the basic of Artificial Intelligence and how it works CLO2: Student can develop of AI and implement them systematically in several simple cases with programming languages, which can be used in modeling and designing applications in the field of information technology. CLO3: Students are able to implement several algorithms used in the field of AI with the programming language they master.
Prerequisite / Corequirisite *(if any)*	-
Name(s) of Academic Staff	Anindya Ananda Hapsari, S.ST., M.IT

Topics and Delivery Hours (F2F)	(see table below)
Assessment Methods	1. Assignment (30%) 2. Quiz (10%) 3. Mid Semester Assessment (20%) 4. Assessment (40%)
Special Requirements / Resources *(Eg Software, Computer Lab, Simulation Room etc)*	-
References	1. Anindya Ananda Hapsari, Belajar Artificial Intelligence dengan Python Bagi Pemula, Media Sains Indonesia, Jakarta 2020 2. AWS, The Machine Learning Journal, 2019 3. Suyanto, Artificial Intelligence, 2021 4. Dicoding. Belajar Machine Learning untuk Pemula, 2024

No	Topic	Hours
1	Introducing of Artificial Intelligence	2
2	History, definition, example of AI	2
3	Definition and Sample of Smart Agent	2
4	Start Programming with Python	2
5	Searching Algorithm	2
6	Natural Language Processing DBA	2
7	Natural Language Processing TTS	2
8	Algorithm AI Implementation NLP	4
9	Machine Learning Model	4
10	Supervised and Unsupervised Learning	4
11	Implementation of Expert System	4
12	Artificial Neural Network	4
13	Sentiment Analysis	4
14	Implementation of AI algorithms to solve the problem.	4
	TOTAL HOURS (F2F)	42